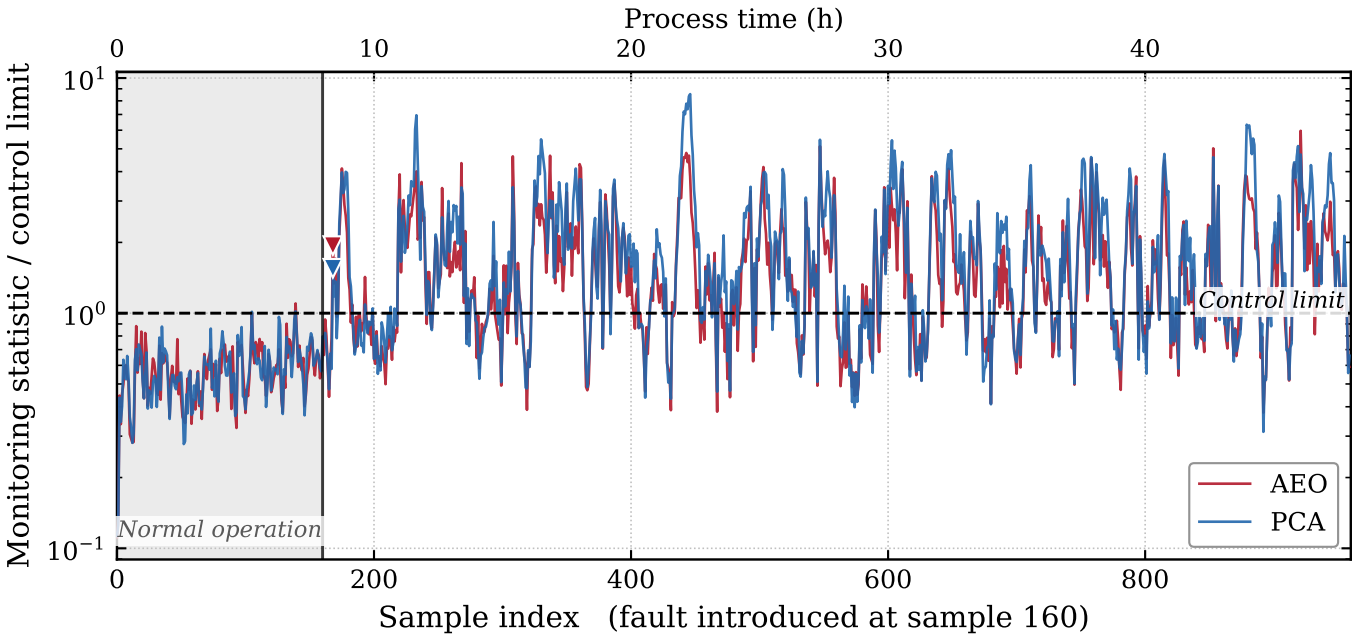
Fault Detection Performance for IDV(11)

Combined T^2 and SPE statistics — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	8.7 samples (26 min)	MDR 0.360	detected in 7/7 runs
PCA	delay	7.7 samples (23 min)	MDR 0.296	detected in 7/7 runs

PCA detects 1.0 samples (3 min) earlier; PCA missed-detection rate lower by 18%